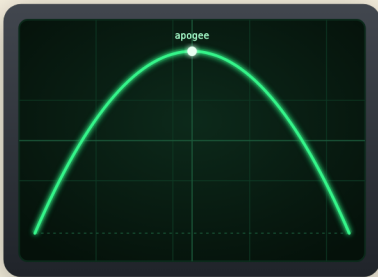


Watch a Cannonball Fly

No chip. No software. This machine answers an ancient, deadly question — *where will the projectile land?* — the very question that gave birth to the computer age.



That green arc is a cannonball in flight — and this machine is figuring out its path right now.

It's an **analog computer**. Instead of counting in 1s and 0s, it uses **electricity to act out the physics**: one voltage is gravity, one *is* speed, one *is* height. It doesn't calculate the flight — it **becomes** it.

1

Gravity = a voltage

A knob sets how hard gravity pulls — a steady push, downward.

2

Add up → speed

An **integrator** totals that pull over time to get the ball's speed.

3

Add again → height

A second integrator totals speed to get height — the arc on screen.

The problem that built computing: for centuries armies asked “aim how high, land where?” Solving a projectile's path — by the bookful — drove the first great computers (ENIAC, 1945, was built for artillery firing tables). From this need grew the whole industry behind everything around you.



Ask the docent to turn the GRAVITY knob. The whole arc reshapes *instantly* — flatten it and the shot sails like it's on the Moon. Nothing is reprogrammed; the wires simply *are* the new equation.

*“It doesn't calculate the motion — it **becomes** the motion.”*